A DESIGN METHOD FOR SMALL SENSOR ARRAYS IN ANGLE OF ARRIVAL ESTIMATION

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ABSTRACT

The use of small sensor arrays in modern signal processing systems has recently become more common due to the increase in computational processing power and interest in intelligent sensing and surveillance. However, not much information is available on the design of small sensor arrays having arbitrary geometry, that effectively can accomplish these tasks. In this paper we address the problem of designing such small sensor array systems for angle of arrival (AOA) estimation algorithms. Two different cost functions are derived and their applicability is demonstrated in simulation. The accuracy of the AOA estimates is also studied for two different array configurations.

1 INTRODUCTION

Angle of arrival estimation has been the interest of research for quite a long time. It is an important field of study for both military and civilian applications such as acoustic detection, surveillance and tracking. Time-delay based AOA estimation has been addressed in several papers. Also some robust methods, that can be applied even when sensor malfunction may occur, have been derived [7, 8, 9]. However, there has been very few papers about small sensor arrays and their design. Design methods have been published in the case of beamforming for both uniform linear arrays (ULA) [1], and nonuniform linear arrays [2], but they cannot be applied in our case.

The paper is organized as follows. In Section 2, the basics of an angle of arrival estimation system are explained. Some general design principles for an AOA system are given in Section 3 along with two different approaches for the design. Simulations using the derived cost functions are covered in Section 4. Finally, Section 5 concludes the paper.

2 ANGLE OF ARRIVAL ESTIMATION

Consider that an acoustic signal $s(t)$ is emitted by the source and received by an array of K sensors. If the sensors are not on the same plane it is possible to uniquely solve both the elevation and azimuth angles of arrival using the differences in the times of arrivals between the sensors [6]. The sensor signals can be denoted as $y_i(t)$, where i is the index of the sensor ranging from 1 to K. The problem now is to find the angle of arrival of the source based on the sensor signals.

In this paper, the number of available sensors is assumed to be relatively small, i.e. ranging from four to eight microphones. Out of these sensors, some can even be broken or give otherwise deteriorated outputs.

There are a couple of possible approaches for solving the problem based on time delay estimation. The first is to use the conventional cross-correlation calculation and use the delays that correspond to the maximum cross-correlation value. These time delays can then be used to calculate an estimate for the angle of arrival. There are several different methods that can be used for the estimation [6, 7, 8]. The second approach is to use a grid search algorithm. This approach uses the fact that for a certain angle there exists a unique combination of delays between the sensors, assuming that all the sensors are not on the same plane [9]. This is utilized by finding the grid point that gives the best match with the received sensor signals.

3 DESIGN PRINCIPLES

The array design can be done by optimizing a cost function that is dependent on the design parameters. These parameters include: number of sensors, probability of sensor malfunction, desired accuracy for estimation for different angles, the accuracy of the time delay estimates, restrictions for sensor placement and frequency content of the signal itself.

The accuracy of the time delay estimation has an effect on the accuracy of the angle of arrival estimate consequently. Some of the parameters also have a direct effect on the accuracy of the time delay estimation.

Another factor that affects the accuracy of the angle of arrival estimation process, besides the accuracy of the time delays, is the placement of the sensors. For example, in the case of two sensors the most accurate estimate for the angle can be achieved when the propagation vector is orthogonal to the sensor vector.

Both of these effects can be seen in Figure 1. The effect of time delay quantization and sensor placement on the angle estimation can be seen in the upper and lower plots, respectively. In this construction a sensor vector with a length of one meter was used, an acoustic source signal traveling at $c = 340$ m/s was assumed, and the sampling frequency was $F_s = 2$ kHz.

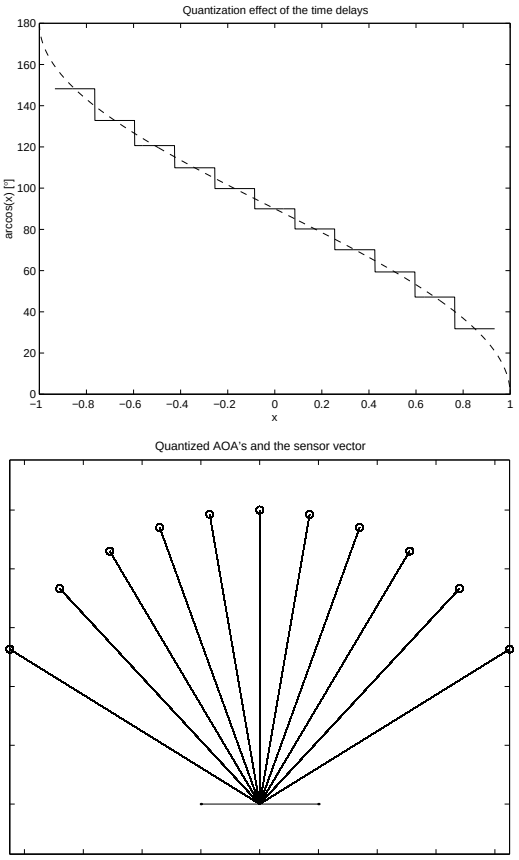


Figure 1: The effect the time delay quantization (upper) and sensor placement (lower) on the angle estimation.

The application dependent factors such as the required size and shape of the array may introduce quite heavy restrictions. The minimum amount of sensors can be derived from the dimension of the angle of arrival estimate, i.e. if both azimuth and elevation angle are needed, then all the sensors must not reside on the same plane. Thus, the minimum amount of sensors in this case is four.

The goals for the performance of the system can usually be stated by the desired accuracy for the AOA estimate at different angles, desired robustness i.e. the amount of sensors that may malfunction and the effect this has on the accuracy of the estimation. The latter may be optimized e.g. by using a weighting scheme.

Two different approaches can be used to model the error in the time delay estimates, which then can be used to construct a cost function that can be minimized. The first approach models the errors in the time delays using quantization, which is a nonlinear process. Thus, the effect of the quantization on the AOA estimate cannot be solved analytically. This leads to a computationally heavy *brute force* method in which a dense enough grid consisting of AOAs denoted by $\boldsymbol{\theta}^{(m)}$ has to be constructed. This can be done by projecting propagation vectors, that represent the direction and speed of the acoustic wave, $\boldsymbol{k}^{(m)}$ to the sensor vectors $\boldsymbol{x}_{ij}$, as follows

$$\boldsymbol{t}^{(m)} = V\boldsymbol{k}^{(m)}, \tag{1}$$

where V is a matrix composed of the sensor vectors [6]. The length of the propagation vector is inversely proportional to the propagation speed c, i.e. $||\boldsymbol{k}^{(m)}|| = \frac{1}{c}$. The time delay vectors $\boldsymbol{t}^{(m)}$ then have to be quantized to the nearest possible value according to the sampling frequency F_s. Now, these quantized time delay vectors can be used to get the quantized propagation vectors $\boldsymbol{k}_q^{(m)}$ corresponding to the AOA estimates $\boldsymbol{\theta}^{(m)}$. This grid can now be determined for different array configurations and the errors for the angle estimate $\tilde{\boldsymbol{\theta}} = \boldsymbol{\theta} - \hat{\boldsymbol{\theta}}$ can be evaluated. Now, a cost function J, can be defined for example as the maximum error

$$J_{max} = \max_m \left(||\tilde{\boldsymbol{\theta}}^{(m)}||\right) = \max_m \left(||\boldsymbol{\theta}^{(m)} - \hat{\boldsymbol{\theta}}^{(m)}||\right). \tag{2}$$

This can then be minimized for a certain class Ω, that contains all the possible parameter sets for the array, by finding

$$\Omega_{opt} = \arg\min_{\Omega} J_{max}, \tag{3}$$

where M is the amount of points in the grid. If the accuracy is more crucial for some angles $\boldsymbol{\theta}^{(c)}$, the cost function for those angles can be weighted by a suitable factor ($w^{(c)} > 1$), which will yield better estimates for the weighted angles.

There are several possibilities to perform the optimization. A thorough description of an optimization scheme called *simulated annealing* (SA) can be found in [3], and it could be used in the optimization here as well. It has already been successfully used for the optimization of both planar and nonuniform linear sensor arrays in [5] and [4], respectively. *Genetic algorithms* (GA) are another approach to solve optimization problems.

The second approach uses a model for the distribution of the error in the time delay estimates. If the estimated time delays are denoted as

$$\boldsymbol{t}_q = \boldsymbol{t} + \boldsymbol{e}, \tag{4}$$

where $\boldsymbol{t}$ denotes the real time delays and $\boldsymbol{e}$ denotes the estimation error. Now, the estimated propagation vector becomes

$$\begin{aligned} \boldsymbol{k}_q &= (V^TV)^{-1}V^T\boldsymbol{t}_q \\ &= (V^TV)^{-1}V^T(\boldsymbol{t} + \boldsymbol{e}) \\ &= \boldsymbol{k} + (V^TV)^{-1}V^T\boldsymbol{e} \\ &= \boldsymbol{k} + \boldsymbol{e}_k. \end{aligned} \tag{5}$$

Using this approximation the covariance matrix of the error in the propagation vector estimate depends only on the sensor vector matrix V and on the covariance matrix of the error in the time delay estimate. Thus,

$$E\{\boldsymbol{e}_k\boldsymbol{e}_k^T\} = (V^TV)^{-1}V^TE\{\boldsymbol{e}\boldsymbol{e}^T\}V(V^TV)^{-1}. \quad (6)$$

Thus, the trace of the covariance matrix can be used as a cost function

$$J_{tr} = \mathrm{tr}(E\{\boldsymbol{e}_k\boldsymbol{e}_k^T\}). \quad (7)$$

The optimal solution for the problem can be found by setting the partial derivatives of the cost function to zero as

$$\nabla J = \nabla \mathrm{tr}(E\{\boldsymbol{e}_k\boldsymbol{e}_k^T\}) = 0, \quad (8)$$

and finding the parameter values that give the minimum cost within the parameter set and taking into account the given restrictions for the design. Depending e.g. on the amount of parameters that are optimized, the gradient in Equation (8) can be simple or on the other hand very demanding.

4 SIMULATIONS

The angle grid used in all of the following simulations was generated by first choosing 1000 elevation angles with equal spacing on a unit sphere. Now, the spacing of the azimuth angles should be done in such a way that the amount of angles at each elevation is in proportion to the circumference of the corresponding circle on the surface of the unit sphere. This way the 3-D-spacing will be well balanced.

This was accomplished by calculating the radius r of the corresponding circle on the unit sphere for each elevation angle. Since the circumference of the unit circle equals to $2\pi \cdot 1$ and the circumference of a circle is $2\pi \cdot r$, the spacing of the azimuth angles at each elevation are in proportion to the radius. Now, all that needs to be done is to define the maximum amount of angles given to elevation angle 0°, and then round the amount of angles for each elevation to an integer and divide 360° with that integer to get the azimuth spacing.

A simple array design task could for example have the following restrictions: a total of four sensors can be used, the sampling rate is 8 kHz, and the array configuration is such that three sensors are placed in a triangular form at the bottom and the fourth sensor must be placed in the center, but the height may vary. The length of each side in the triangle is 0.1 meters and the speed of the acoustic wave is assumed to be 340 m/s.

When the height of the fourth sensor is varied from 3 centimeters up, and the cost function in Equation (2) is used, the curve shown in Figure 2 is obtained.

On the other hand, when the cost function in Equation (7) is used, the resulting curve is quite similar to the previous one, as illustrated in Figure 3. Here the covariance matrix of the noise is assumed to be

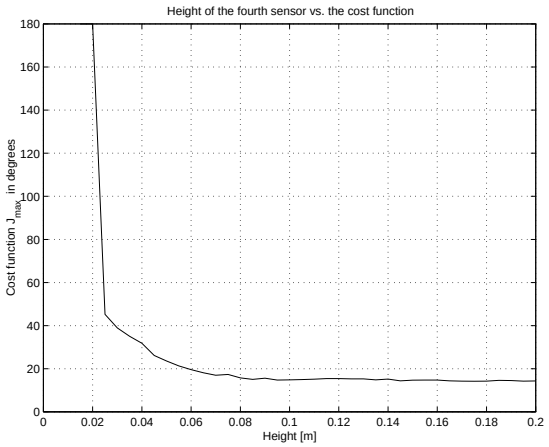


Figure 2: Cost function J_{max} as a function of the height of the fourth sensor.

$$E\{\boldsymbol{e}\boldsymbol{e}^T\} = \sigma^2 I, \quad (9)$$

where σ^2 denotes the noise variance and is assumed to be unity. Now, the cost function in Equation (7) can be simplified to

$$J_{tr} = \sigma^2 \mathrm{tr}((V^TV)^{-1}). \quad (10)$$

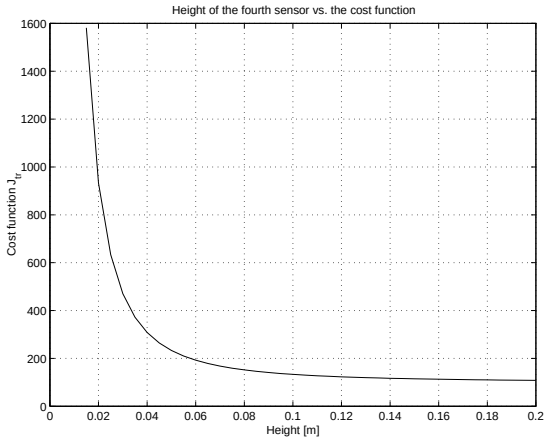


Figure 3: Cost function J_{tr} as a function of the height of the fourth sensor.

As can be seen from the figures, the cost seems to decrease asymptotically towards a certain bound, and it would be quite reasonable to use a tetrahedron shape ($h = 0.1 \cdot \sqrt{2/3}$) in order to save space.

In this case, the locations for three of the sensors are fixed, thus only the location of the fourth sensor needs to be optimized and a simple search can be used to find

the optimum solution. However, if the problem is more complex and there are several parameters to be optimized, a more efficient solution is needed.

To evaluate the accuracy of the estimated angles, the amount of error is plotted as a greyscale value in Figures 4 and 5.

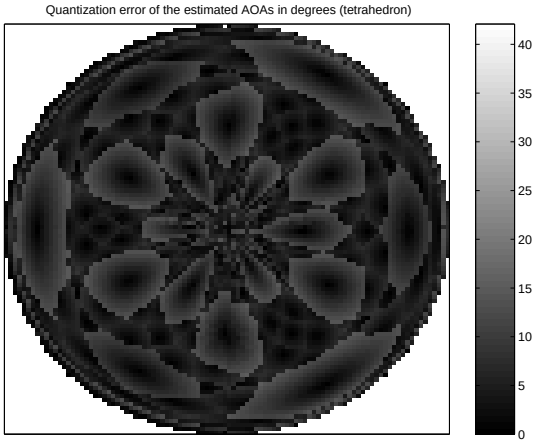


Figure 4: The quantization error of the AOA estimate using a tetrahedron shaped sensor array.

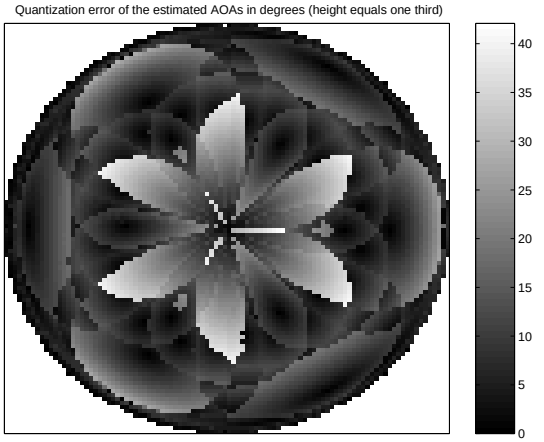


Figure 5: The quantization error of the AOA estimate using 1/3 of the height for fourth sensor.

The figures were formed by projecting propagation vectors corresponding to the AOAs into the xy-plane, but the distance of the points from the origin is scaled with respect to the original z-coordinate for illustration purposes, otherwise most AOAs would be near the edge of the circle. As can be seen, a poor design can severely deteriorate the accuracy of the AOA estimation, and introduce heavy quantization in the AOAs as well.

5 CONCLUSIONS

The problem of designing small arrays for AOA was addressed in this paper along with a method for the optimization of the accuracy of the angle estimate. Some design principles were also given, but the optimization routines themselves were left out, since any general scheme, e.g. simulated annealing or genetic algorithms, can be used for the task. Two different cost functions were given for the optimization task, and their applicability was demonstrated in simulation. Also the accuracy of the angle estimates was studied for two array configurations.

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